Career Summary – Bach Le

Bach has built a national and international reputation as a researcher in Software Engineering, particularly Automated Software Debugging. He has fostered leadership and excellence in teaching and learning, mentorship, and research, including industry engagement in research and teaching.

Research and Training

- Among the most well-recognized researchers working on Automated Software Bug Fixing as evidenced by high-quality publications in the research field of Software Testing and Bug Fixing with an **h-index of 15 and i10-index of 18 (total 1649 citations): in total 32 publications**, including 1 book chapter, 3 journal papers and 28 conference papers at prestigious and top venues in Software Engineering.
- Research funding:
    - Over $500K funding as sole investigator, including:
        - DECRA'22: $488K awarded from the DECRA funding started from January 2022.
        - Early Career Researcher Award from CIS's Computer Science group in 2022: $20K
    - Collaborator of ~$2M research fundings that I co-wrote with Professor David Lo (Singapore Management University) and Professor Corina Pasareanu (Carnegie Mellon University). The fundings were awarded to the partners who are lead investigators, not the University of Melbourne.
- Industry Engagement and Impact:
    - Represented the University of Melbourne as a mentor at Google Summer of Code (GSoC) in 2019 and 2020. Each student working on a GSoC project with me was awarded ~$5000 USD for successful completion of the project.
    - Delivered invited talk at a Japanese company, namely Cinamon AI, and collaborated with a research team of 4 researchers from the company on automated program synthesis. The collaboration resulted in a product used in the company.
- Recognition:
    - Invited to serve on an expert panel of world-leading researchers on Automated Bug Fixing to discuss future directions for the research area (organized with ICSE 2023 in Melbourne)
- Research Training: Supervising three (3) PhD students as primary supervisor.
    - Thanh-Dat Nguyen (since February 2021, full time): first-authored papers published with me include 1 x ICSME, 1 x ICSE.
    - Cong-Thanh Le (since February 2023, full time): first-authored papers published with me during PhD include 1 x TSE. Papers published with me before PhD include (first-authored) 1 x ICSE, 1 x ESEC/FSE, 1 x TSE, 1 x ISSRE, and (second-authored) 1 x ESEC/FSE, 1 x ICSME, 1 x ASE. Cong-Thanh won the prestigious **Google PhD Fellowship and Best Undergraduate Thesis** under my supervision.
    - Vinay Kabadi (since January 2020, part time). Vinay has one paper accepted to ICSME'23 which won **Distinguished Paper Award**.

  In summary, my students are progressing well on their PhD even though they have all been affected by the COVID-19. I joined the University in July 2019, and shortly after that is the COVID-19 which delayed the PhD recruitment process.

Teaching and Learning

- Led 40 master's students (~10 students each team) to engage with industry partners. Two of the teams I supervised **won the Endeavour awards** at the University level for their projects working with the industry partners.
- Graduated three master's students (Master of Research), two of whom were under my primary supervision.
- Involved in teaching and coordinating a wide range of subjects (7 subjects) from undergraduate to graduate levels, with class sizes ranging from 45, 200-300, and 600 students, with an average performance of 3.97/5.0 across the subjects, of which the compiler theory subject that I taught received an outstanding 4.4/5.0. My overall performance of 3.97/5.0 exceeds the requirement of 3.80/5.0 set out by the University.
- Awarded $10K by the University for Models of Computation - COMP30026 on BSL learning and dual delivery. The money was well spent on producing a series of short tutorial videos/interactive dialogues that were well-appreciated by students.

Service and Leadership:

- Regularly invited as PC member/reviewer of various prestigious conferences and journals such as ICSE, ASE, ISSTA, TSE, TOSEM, TNSM.
- Invited as an expert speaker among an expert panel of four world-leading researchers to discuss the future of automated software bug fixing at Automated Program Repair, collocated with ICSE'23.
- Served as chair for:
    - Student Research Competition Track, ASE'22
    - Demonstration Track, IBC'22
    - Software Engineering Track, SoICT'23
- Served on hiring committee of the University of Melbourne in 2022 for hiring a lecturer/senior lecturer position in CIS.
- Served on DECRA's discussion panel at CIS to share my experience on writing a successful DECRA application. I also personally shared my DECRA proposal, helped my peers in providing comments and suggestions in improving their proposals and responses.
- Mentored my colleague Dr. Van-Thuan Pham (lecturer at CIS) on his DECRA application. Dr. Pham's has then been successfully awarded the DECRA'23.

Career Trajectory.

After a year working as a postdoctoral researcher at Carnegie Mellon University (NASA Research Park, California, USA), I was appointed as a continuing lecturer at the University of Melbourne in July 2019. Over the past four years, I have gradually been developing a strong research track record as evidenced **by 32 publications in total, h-index of 15, i10-index of 18, total 1635 citations, and being awarded a prestigious funding DECRA'22 worth ~$500K**. At the same time, I have developed teaching skills that allow me to interact and engage with students in effective and meaningful ways. Despite the effect of COVID-19 during my early years at the University of Melbourne, my performance in the last four years remains consistently meeting/exceeding expectations as acknowledged by my supervisor. With strenuous effort spent in 2021, I was awarded the DECRA'22 (Discovery Early Career Researcher Award) **in which I was the only candidate in Australia in the area of Software Engineering and Programming Language awarded the funding**. Coming out from the COVID-19, in the year 2022, my research performance was acknowledged by my manager as outstanding/stellar (seven publications at top tier venues, of which four publications are led by my students).

The remainder of this document provides the evidence and further information for my career.

<u>Research and Research Training</u>:

Research Summary.
My main research theme revolves around automated software analysis and empirical studies, with a special focus on automated software testing and bug fixing. I have published 32 publications in the above research areas. My most cited research paper "History Driven Program Repair", which I first authored, has attracted 366 citations, and was adopted by the industry (i.e., the GetAFix bug repair tool by Facebook). GetAFix marks a breakthrough in automated bug fixing, as it is the first ever bug fixing tool successfully deployed in the industry, particularly on Facebook's codebases. The underlying technology of GetAFix relies on bug fixing patterns mined from software repositories, a technique that has been directly inspired by "History Driven Program Repair" (as explicitly mentioned in GetAFix's paper). The "History Driven Program Repair" paper remains the most cited paper from the International Conference on Software Analysis, Evolution, and Reengineering (SANER) since 2016.

Automated debugging with a special focus on Automated Program Repair: Ten years ago, automated software bug fixing remained a futuristic idea. This has gradually materialized via several active research projects and breakthroughs in the field, among which my research has made a strong impact and influence. Following on the success of the "History Driven Program Repair" paper, I continue to push the boundary of automated bug fixing further with recent works that I published at top tier conferences and journals. Being awarded the DECRA'22 funding worth ~$500K AUD as sole PI on automated bug fixing, I leveraged this opportunity to expand my network of research, hiring students, etc. This resulted in a stellar year 2022 with seven publications at ICSE, ESEC/FSE, ISSTA, TSE, of which four papers are led by my PhD students.

My research has been gradually built up from two orthogonal dimensions: empirical and human studies, and automated tools/techniques informed by actionable insights learned from the studies. Together with my students, I built systems that involve mining massive amounts of software artifact data, such as code bug fixes, studying human perceptions on software development, such as code correctness and comprehension, etc. Building up from insights extracted from those studies, my research offers better understanding of the nature of research problems that we tackle, identifies areas that can be improved and how to achieve them via automated techniques that leverage syntactic and semantics analyses of software programs. Particularly, in addition to the syntactic search such as mining software repositories for syntactic bug patterns used in "History Dirven Program Repair" and the likes, my research also leverages semantics program analysis such as symbolic execution, constraint solving, program invariants, etc. I believe that the synergy between empirical/human studies and the combination of both syntactic and semantics analyses to automated software debugging in general is vital and worthwhile to pursue further.

Exploration of research areas related to automated debugging: I also recently extended my research focus to other related areas such as software security and dependable machine learning (e.g., verification of neural networks). My PhD student Thanh-Dat Nguyen has achieved encouraging initial results in these areas that have been accepted to top tier conferences in Software Engineering such as ICSE and ICSME. Together with my students, we explored the use of Software Engineering techniques such as invariant inference on inferring properties of Graph Neural Networks (GNNs). We showed the effectiveness of such approaches and their numerous applications on ensuring safety and correctness of GNNs such as using inferred properties to effectively detect and defend against attacks on GNNs. In the near future, I look forward to closing the gaps between multiple research fields such as automatic bug detection and repair of machine learning libraries and models by leveraging my expertise in the area of automated software repair.

To realize the above research directions, I have been collaborating with world-renowned professors in the field of Software Engineering, both within and outside Australia such as Dr. Xin Xia (Director of Software Engineering and Application Technology Lab, Huawei. Previously, Lecturer at Monash University until 2021), Professor Gary T. Leavens (University of Central Florida), Associate Professor Corina Pasareanu (Carnegie Mellon University and NASA Ames), Associate Professor Claire Legoues (Carnegie Mellon University), Professor Koushik Sen (University of California, Berkeley), Professor David Lo (Singapore Management University), Professor Willem Visser (Senior Principal Applied Scientist at Amazon). At the University of Melbourne, I am co-supervising PhD students with Associate Professor Toby Murray, Dr Christoph Treude (Senior Lecturer), and Dr Patanamon Thongtanunam (Senior Lecturer).

Quality and Productivity.
My research record shows a continuing standard of excellence, in particular, the last four years I have excelled to become a strong researcher in my field. This is demonstrated by my publication record, with 31 conference and journal articles in total that attracted 1649 citations (h-index of 15 and i10-index of 18), and over $500K in funding as sole PI since my appointment at the University of Melbourne in July 2019. I regularly publish my research papers at top tier and good venues in Software Engineering, including ICSE, ESEC/FSE, ASE, ISSTA, TSE (journals), ICSME, ICST, and ISSRE These venues generally have acceptance rates at around 20-25%.

Research Funding.
Since my appointment at the University of Melbourne in 2019, I have secured both internal and external fundings of approximately over $500K as sole PI. In addition, I have also gained experience of applying for funding from the industry (Facebook funding) (not selected but was finalist). I have also co-written proposals with my collaborators which received in total ~$2M AUD (awarded to the collaborators, not the University of Melbourne), which allow us maintain our close collaborations.

In particular, my engagement on research fundings is listed below:

- Sole PI fundings at UoM: **Total over $500K AUD**
    - **External Funding:**
        - ARC DECRA 2022. Title: *Practical Automated Software Bug Fixing via Syntactic and Semantic Analyses*.
            - Awarded amount: $488K AUD. Sole PI.
    - **Internal Funding:**
        - ECR grant for early career researcher awarded by CS group at CIS, 2022. Title: *REFIXAR: Practical Automated Repair of Software Regression Errors*.
            - Awarded amount: $20K AUD. Sole PI.
        - Seed grant awarded by CS group at CIS, 2023. Title: *Formally Verified Syntax- and Semantics-guided Automated Proof Repair*.
            - Awarded amount: $8K AUD, Co PI with Christine Rizkallah (Senior Lecturer at CIS)
- Co-written (as collaborator) fundings (awarded to other universities not UoM): **Total ~$2M AUD**.
    - Ministry of Education, Singapore, 2022. Title: *Uncovering Vulnerabilities in Machine Learning Frameworks via Software Composition Analysis and Directed Grammar-Based Fuzzing*.
        - Awarded amount: $500K SGD. Funded to Singapore Management University, in collaboration with Professor David Lo
    - NSF, USA, 2019. Title: *HUman-Guided hybrid Software testing and analysis (HUGS)*
        - Awarded amount: $1M USD. Funded to Carnegie Mellon University, University of California Berkeley, University of California Santa Barbara, in collaboration with Professor Corina Pasareanu, Professor Koushik Sen, and Professor Tevfik Bultan.

- Finalist funding (not awarded):
    - Facebook Research funding, 2020. Title: *Agent-based User Interaction Simulation to Find and Fix Integrity and Privacy Issues*.
        - Aimed for $50K USD. Not selected but was chosen as finalist. In collaboration with Assistant Professor Tian Yuan (Queens University, Canada).

Publications.
Since I started my research career in 2013, I have published in total 32 publications, including 1 book chapter at Automatic Software Verification, 3 x ICSE, 4 x ESEC/FSE, 3 x ASE, 3 x TSE, 1 x EMSE, 2 x ISSTA, 4 x ISSRE, 5 x ICSME 1 x SANER, 1 x ICST, 1 x FoSSaCS, and others at smaller venues. Since joining the University of Melbourne in July 2019, I have published 17 papers, including 2 x ICSE, 2 x ESEC/FSE, 3 x TSE, 1 x ISSTA, 3 x ISSRE, 1 x ICST, 2 x ICSME, and 1 x FoSSaCS, and others at smaller venues. Out of these 17 papers, 6 papers are led (first-authored or co-first authored) by my students who are under my primary supervision (PhD) at the University of Melbourne.

In Software Engineering, prestigious journals include Transactions on Software Engineering (TSE) and Empirical Software Engineering Journal (EMSE). Prestigious conferences in Software Engineering include International Conference on Software Engineering (ICSE), The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), International Conference on Automated Software Engineering (ASE), The ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA). The acceptance rates at these conferences often hinge at approximately 20%. Other top venues in specialized areas such as software maintenance, testing and verification include The International Symposium on Software Reliability Engineering (ISSRE), EEE International Conference on Software Testing, Verification and Validation (ICST), The International Conference on Software Maintenance and Evolution (ICSME), International Conference on Software Analysis, Evolution, and Reengineering (SANER), International Conference on Foundations of Software Science and Computation Structures (FoSSaCS). The acceptance rates at these conferences often hinge at approximately 25%.

Impact.
My research impact is partly demonstrated via my publication record (total 1649 citations, h-index 15 and i10-index 18). Overall, my research addresses two important angles in software engineering that have been well-acknowledged by the community, including: empirical software engineering and practical techniques and tools on the adoption of syntactic and semantics analyses of software.

My most cited paper "History Driven Program Repair" (366 citations) demonstrates the practicality of automated bug fixing via data mining, which subsequently is adopted in industry (Facebook's first ever automated bug fixing tool, namely GetAFix). The paper remains the most cited paper at SANER since 2016. My work published at ESEC/FSE in 2017, namely "S3: Syntax- and Semantic-Guided Repair Synthesis via Programming by Examples" (226 citations) was among the first techniques and tools that scales semantics analysis of large real-world programs (up to 237K lines of code) that can be applied to automated bug fixing. My recent work on human study, namely "Smart Contract Development: Challenges and Opportunities" published at TSE in 2019 remains the top 1% of web of science highly cited papers (365 citations).

International Recognitions.
As a well-recognized expert in the field of Automated Software Bug Fixing, I have been invited to talk at various venues, including:

- Invited talk on: "On Reliability of Patch Correctness Assessment", NASA Ames, Mountain View, California, 2019

- Invited as an expert panel to discuss on future directions of automated bug fixing, Automated Program Repair workshop, collocated with ICSE 2023, Melbourne, Australia

I have also been invited to serve as on program committee and reviewer for prestigious conferences and journals in Software Engineering such as TSE, TOSEM, EMSE, ICSE, ASE, ISSTA, etc.

International Collaborations.
Since joining UoM, I have developed active research collaborations not only within Australia, e.g., with Monash University, but also with top researchers in the USA and other countries, e.g., CMU, University of California Berkeley, etc. These collaborations have resulted in research papers at top journals and conferences in the field of software engineering, particularly software testing and bug repair. I have been working with Dr Xin Xia (previously a Lecturer and ARC DECRA Fellow until 2021) at Monash University on automated program repair and have co-authored conference and journal papers. I have known Dr Xin for several years via my PhD supervisor, Professor David Lo at Singapore Management University. I have also been working with Associate Professor Zhenchang Xing, Australian National University, via a collaboration with Prof David Lo. My international collaborations remain strong and active. I have helped my advisor, A/Prof. Pasareanu at CMU, with writing a research proposal recently accepted by the National Science Foundation (NSF), USA. The wonderful support from NSF allows me to continue the strong collaborations with CMU and other partner institutions such as University of California, Berkeley (Prof. Sen) and University of California Santa Barbara (Prof. Bultan). I have also been invited to speak about my research on automated software bug fixing at NASA Robust Software Engineering group in NASA's Ames Research Center, Mountain View, California. In summary, my research collaborations have provided excellent opportunities to broaden my research connections across various sectors, e.g., governmental sectors such as NASA and software companies such as Google (via Google Summer of Code that I served as a mentor).

Research Training.
Since my appointment at the University of Melbourne in July 2019, I have been supervising three PhD students under **my primary supervision** and graduated three master's students.

Ability to attract and retain talented students. As an academic, it is important to develop a good network to attract talented students. Over the years, I have developed strong relationships with my collaborators from Vietnam to bring in talented students overseas. Particularly, I have been working with Professor Quyet-Thanh Huynh, the president of Hanoi University of Science and Technology (HUST) – the best university in Vietnam in computer science. I have been collaborating with Prof. Huynh on several research projects in order to train young talented undergraduate students from the university and offer them research positions at The University of Melbourne after they graduate. This fruitful collaboration not only results in collaborative publications with the students, but also helps bring in several talented students and retain them at The University of Melbourne. Below, I provide details about my current PhD students, two of whom I have had the opportunity to train since their undergraduate study at HUST.

My PhD students are:

- Thanh-Dat Nguyen: Full time, joined since February 2021. Leave of absence from January 2022 to July 2022 due to COVID-19. Co-supervised with Dr. Patanamon Thongtanunam (Senior Lecturer at CIS)
  Papers with me: 1 x ICSME (2022, first authored), 1 x ICSE (2022, first authored), 1 x ASE (2023, second authored).
  Papers submitted (already completed): 2 x ICSE 2023 under submission.
- Cong-Thanh Le: Full time, joined since February 2023. Co-supervised with A/Prof. Toby Murray at CIS. Cong-Thanh recently won **Google PhD Fellowship** under my supervision for his PhD thesis on Automated Program Repair.

Papers with me before PhD: 1 x ISSRE (2021, first authored), 1 x ICSE (2022, co-first authored), 1 x ICSME (2022, second authored), 1 x ESEC/FSE (2022, first authored), 1 x ICSE (2023, co-first authored).
Won **Best Undergraduate Thesis Award** under my supervision at HUST.
Papers with me during PhD: 1 x TSE (2023, first authored), 1 x TSE (2023, second authored), 1 x ASE (2023, third author).
Papers to be submitted (already completed): 1 x TSE (under minor revision), 1 x TSE (rejected, will submit again soon).

- Vinay Kabadi: Part time, joined since January 2020. Leave of absence on various occasions due to COVID-19. Co-supervised with Dr. Patanamon Thongtanunam and Dr. Christoph Treude (Senior Lecturers at CIS)
  Papers with me: 1 x ICSME (2023, first authored). **Won Distinguished Paper Award**.

Although my students have all been affected by the COVID-19, they are progressing well under my primary supervision. Thanh-Dat has papers accepted at top venues in Software Engineering (ICSE, ICSME) and is having two other works in the pipeline that have been submitted in August 2023. Cong-Thanh although joined just in February 2023, has been working with me since 2019 and had several papers at top tier venues with me such as TSE, ICSE, ESEC/FSE, ICSME, and ISSRE. Vinay joined as a part time student since 2020 and has been affected by the COVID-19 during early years at the University. However, Vinay had a paper accepted to ICSME in 2023 which won Distinguished Paper Award, and another paper halfway through obtaining initial encouraging results.

I have also supervised/co-supervised three master's students (by research) and all of them have graduated. Apart from research training from UoM, I also actively engage with the industry to supervise students doing research that is of interest to the software industry. Particularly, I served as mentor for Google Summer of Code in 2019 and 2020, supervising students working on the Java PathFinder project – a project on automated program analysis developed by NASA Ames research center.

Top Five Recent Publications.

- "Smart Contract Development: Challenges and Opportunities", published at Transactions on Software Engineering (TSE), 2019. This work attracted 273 citations since 2019 and remains the top 1% of web of science highly cited papers. This work was the first human study to explore challenges and opportunities in smart contract development in blockchain.
- "On Reliability of Patch Correctness Assessment" published at 41st ACM/IEEE International Conference on Software Engineering (ICSE), Research Track (12 pages), 2019. This is the first human study that explores how the correctness of machine-generated patches should be assessed to allow fair comparisons of program repair techniques. The paper has attracted 68 citations since 2019.
- "Invalidator: Automated Patch Correctness Assessment via Semantic and Syntactic Reasoning", published at Transactions on Software Engineering (TSE), 2023. This work combines syntactic and semantics reasoning of software programs to automatically identify correctness of patches generated by automated bug fixing techniques.
- "Chronos: Time-Aware Zero-Shot Identification of Libraries from Vulnerability Reports" published at the 45th IEEE/ACM International Conference on Software Engineering (ICSE), Research Track (12 pages), 2023. This work uses machine learning techniques to automatically identify possible vulnerability in software library.
- "AutoPruner: Transformer-based Call Graph Pruning" published at the 21st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Research Track (12 pages), 2023. This work proposes to use machine learning

techniques to automatically construct precise call graphs in software programs, allowing to analyze semantic behaviors of programs more precisely.

Teaching and Learning:

Philosophy. Teaching is an important part of knowledge transfer and discovery. It not only equips learners with skills needed for their future career, but also strengthens their ability to independently develop and pursue their own interests. Such a process is, however, not solely a one-way interaction from teachers to learners, but instead, an exciting collective whole of reciprocal improvement of both. As a researcher in Computer Science - a core to fast-paced evolution of technologies, I enjoy engaging in this intriguing process to nurture young talents in the field via teaching a wide ranges of subjects at the University of Melbourne, ranging from core undergraduate subject such as Models of Computation (600 students) to graduate subjects that are specialized in computer science theory such as Programming Language Implementation (Compiler theory, 45 students).

Performance. I have been involved in teaching seven subjects, of which my average teaching performance is 3.97/5.0 overall. My teaching performance exceeds the requirement of 3.8/5.0 set out by The University of Melbourne. I describe in detail the subjects that I taught below:

- *SWEN20003 Object-oriented Software Development*. **Role: Coordinator**, undergraduate level, ~400 students. **Performance**: 4.1/5.0 on average. Especially received 4.32 on clear expectations and assessment, and 4.17 on very good learning experience on the subject.
  Contributions and innovations: I implemented new approach on my teaching for the subject, including: (1) providing several individual meetings with students (on demand) outside office hours to guide students through their knowledge on the subject and future career, (2) engaging more interactivity with students via live demonstrations in lectures, which have not been done in this subject before. I showed students interactive debugging to progressively reason about errors in programs, (3) encouraging further interactivity and engagement from students by rewarding best student projects with prizes.
- *COMP90045 Programming Language Implementation*. **Role: Co-lecturer**, graduate level, ~45 students. **Performance**: 4.4/5.0 on average. Especially, received 4.8 on clear assessment requirements, and 4.6 on satisfaction with learning experience, study materials, etc.
  Contributions and innovations: I taught 50% of this subject on compiler theory. Together with the coordinator of the subject, I designed the project assignment which requires students to implement a complete compiler for a toy programming language that we designed for this subject. To facilitate teaching this subject, I implemented the compiler myself to self-teach on the concepts that I need to deliver to students. The project for this subject must change every year to avoid students copying compilers developed by students in previous semester. To facilitate the changes, we introduced new features in the language, e.g., arrays, pointers, etc to enforce new implementation in the compiler. Although the amount of work needed to learn the theory and implement the compiler is enormous for this subject, the students' overall satisfaction was high.
- *COMP90082 Software Project*. **Role: Co-lecturer**, graduate level, ~130 students. **Performance**: 3.9 on average. Especially received 4.35 on useful interaction and engagement opportunities, and 4.0 on useful feedback on progress.
  Contributions and innovations: This is a special subject that moves every year since each year we need new industry projects for master's students (working in groups of 10 students each) to work on. As a result, I worked with the coordinator of the subject to manage the projects in a way that each project is of equal good standard for students to work on. As part of the effort in achieving

high quality project and supervision, I worked on hiring project supervisors from the industry, communicating with the supervisors on assessment criteria and monitoring groups' progress via weekly meeting with supervisors.

- *COMP30026 Models of Computation*. **Role: Coordinator**, undergraduate level, 600 students. This subject is very special in that the coordinator of the subject Professor Harald Sondergaard unexpectedly retired early. As a result, the department asked me for a helping hand, and I thus volunteered to drop my teaching allocation on COMP90045 (45 students) to take over COMP30026 (600 students). It was a tremendous effort for me to help the department deal with this subject. I have now handed it over to other colleagues who have more suitable background to teach the subject. I taught this subject over two semesters as outlined below.
  Performance:
  - Semester 1 2021: 3.85 on average. Especially received 4.0 on clear assessments and 4.0 on helpful resources and materials.

    Contributions and innovations: We adopted several new approaches in this subject when I took over. We received $10K AUD in total of funding from the faculty to facilitate blended active learning. We used the funding to hire tutors to create interactive short dialogues, in which two tutors informally discuss their views over the content of the lectures each week. A tutor acts from the view as someone who is new and curious to learn about the knowledge in the subject, while the other tutor acts as an experienced teacher in the subject to explain key concepts/takeaway. A remark on these interactive dialogues is that they were very well received by students and these dialogues continue to serve the subject very well in several next semesters. Other changes and innovations introduced in this subject are that we employed the online learning platform Grok for implementing assignments and final exam. This allows students to have hands-on practical coding experience to implement the theory they learned in the subject. Also, this allows the teaching team to implement **automated assessment** to more easily mark and provide concrete feedback to students as to where the students have failed and can improve. This automated assessment approach continues to serve the teaching team very well in terms of marking and providing feedback to student's solutions.

  - Semester 2 2022: 3.6 on average. Note: response rate was very low (60 out of 600)

    Contributions and innovations: This semester, in order to accommodate a new lecturer into this subject, I volunteered to switch my teaching part from the previous semester (logic) to a new part (finite state automata). As a result, this semester was very heavy for me to learn and teach the new part that I was not much exposed to before, which was outside my research expertise. Despite the difficulty, I regularly improve my knowledge in the area by reading several research papers in the field and adopted some of the advanced concepts to the subject. For example, I realized that the subject so far has only taught students generic methods to prove program termination, but not on proving program non-termination. Realizing the gap, I adopted changes to the lectures to provide a more complete view to students on both methods to prove termination and non-termination. The result of this is that students acknowledge the two sides of the coin on the long-standing termination problem in computer science.

- SWEN90013, SWEN90014, and SWEN90009 over various semesters in 2020 and 2021. **Role: Project supervisor** that supervises master's students doing projects for industry partners. **Performance**: performance score is not available as I was the supervisor, not co-lecturer. Under my primary supervision, two (out of four) of my teams have **won University Endeavor Awards** for their innovative projects on blockchain-based payment system and reusable coffee cup.

What I Can Offer to Teaching at Your Esteemed Institution. I have gained tremendous teaching experience over four years at The University of Melbourne, teaching from undergraduate subjects to graduate subjects with class sizes from 45, 130, 400, to 600 students, and from theory to practical subjects. I believe that my teaching experience can offer a diverse range of knowledge at your esteemed institution.

For example, I am comfortable teaching medium graduate subjects such as Software Project that involves working with industry partners in which students are given opportunities to work with real projects in the industry. The challenges underlying these subjects include: (1) How to find adequate number of projects from industry partners for students to work on, (2) How to ensure that the projects are of equal standard to maintain equality among students, and (3) How to find suitable supervisors from the industry to help supervising students and giving students advice in a timely manner. To overcome these challenges, it is critical to have a wide network of industry partners to begin with. This allows us to find suitable projects to choose from and also to find supervisors from the industry who are willing to supervise students on the projects. With the networks that I have established via teaching this subject at The University of Melbourne, I am confident that I can teach well and bring in fresh perspective and industry connections for the graduate subjects such as Software Project and so on.

I am also comfortable with teaching theory subjects such as Compiler Theory. I believe that my experience in teaching theory subjects in core computer science would bring more flexibility to complement colleagues at your esteemed institution.

I always make myself ready whenever my institution needs a helping hand, e.g., as I have discussed on my willingness to drop teaching allocation of a subject with 45 students to help take a subject with 600 students, which is more than 10 times the size of the class that I was originally assigned to teach. I believe that my willingness and readiness offer additional flexibility to teaching allocation at your institution.

Approaches to Teaching.

"*Curiosity is the wick in the candle of learning*" – William Arthur Ward

To demonstrate the quality and innovation in my approaches to teaching and learning, I focus on the following strategies: *problem-based learning*, *peer learning* and *curiosity driven learning*. The nature of the subjects that I taught allows me to implement these approaches in a way that theory and practices are very well-blended, offering students both hands-on experiences and the foundations needed to further pursue their own curiosity driven learning, e.g., going from the theory of computations to the compiler theory and implementation. I explain my approaches in detail below.

"*Boredom will always remain the greatest enemy of school disciplines*" – Fritz Redl

Lectures alone in classroom-based teaching of presenting theory followed by examples may not sufficiently inspire students in their learning. Thus, apart from lectures, I always look for opportunities to encourage students to learn with their peers in an iterative process via problems that cover various topics in the lectures. Such opportunities include individual assignment problems and project-based group assignments. In individual assignments, I always look for individual problems that have increasing difficulty, covering topics discussed in lectures and often one question that is advanced and requires students to think out of the box. Although it's individual assignments, I always encourage students to work with their peers and come up with different solutions on their own. This way students learn to work with and receive feedback from their peers. The advanced question included in the assignment, which may contain knowledge not covered in lecture, often is used to trigger students' curiosity about what the knowledge learned in lectures can be

stacked together to solve more difficult problems. For group-based projects, I always look for problems that are practical and needed in the software industry. This way the group-based projects can give students hands-on experiences of working in a simulated environment of the software industry. Like individual assignments, in group-based projects, students always have opportunities to work with and receive feedback from their peers. Along the way, the projects are broken down to several milestones, allowing students to be able to better control their progress, have timely feedback for each milestone to ensure that they will finally deliver the project in a timely manner with good quality.

To assist students with problem-based learning, apart from lectures, I and my colleagues have also been awarded $10K budget by the University on creative learning. We created short interactive dialogues in which two people informally discuss related topics to summarize lectures from different perspectives. The conciseness and interactivity offered in the dialogue help summarize key ideas that are helpful for students to understand and remember the takeaways. This approach has been much appreciated by students (as indicated in students' comments over several semesters).

"*The best practice is inspired by theory. The best theory is inspired by practice*" – Donald Knuth

I approach teaching with the following three key deliveries: *What* – introduce a problem, *Why* – why it is important, and then *How* – how to solve it. To connect these pieces, I always strive to blend theory and practices together to offer students a more complete view of both theoretical foundations and what the theory can be applied on to solve real problems in practice. This way students learn to remember the theory while appreciating their practicality. Sometimes, to help students in more easily digesting the theory, I use analogies in real life as examples to illustrate the ideas.

For subjects that require open-ended discussions such as project-based subjects, it's critical to equip students with analytical reasoning, persistence, independence, and teamwork. To equip students with those skills, as a supervisor, I work with students at a lower level at the beginning, and subsequently let the students' creativity guide their way when they have acquired sufficient background. This way, students are given a chance to learn experiences from the advisors, while being able grow independently. Following this methodology, I have successfully supervised four groups of master's students (~10 students per group) in accomplishing their projects working with real industry partners. Two out of the four teams subsequently went on to win the Endeavour Awards based on their project work at the University level.

Conclusion.

In short, I enjoy teaching as a lecturer and mentor. I try to be a prepared, interactive, and mindful teacher to keep students engaged and encouraged. I believe that these are key to effective teaching and mentoring. I hope to have many more opportunities to serve as both roles to nurture young talents in the future, and I am confident that I am well prepared to do so.

Service and Leadership:

Service and Leadership in the University.

Hiring committee at CIS. In 2022, I served on the hiring committee of the University of Melbourne for hiring a lecturer/senior lecturer position in CIS, led by A/Prof. Toby Murray at CIS. The hiring process

later on resulted in successful hires such as Dr. Christoph Treude (Senior Lecturer at CIS), Dr. Mansoureh Zahedi (Lecturer at CIS).

Mentoring peers at CIS on DECRA proposal. Following on my successful DECRA in 2022, I have served on a discussion panel at CIS to share with my peers my experience in writing a good proposal for DECRA. I have helped my peers in providing comments and suggestions on improving their proposals and responses. Especially, I have mentored my colleague Dr. Van-Thuan Pham (lecturer at CIS) on his DECRA application. Dr. Pham's has then been successfully awarded the DECRA'23. With mentoring Dr. Pham's proposal, I read Dr. Pham's proposal in detail, and provide comments and suggestions on how it can be improved. Upon receiving comments from DECRA reviewers, I once again read through Dr. Pham's reviews and response to provide comments and suggestions on how Dr. Pham's response to the reviews can be improved. I was extremely happy and felt helpful to see the successful outcome of Dr. Pham's proposal at the end.

Service and Leadership in the Community.

PC member/reviewer of conferences and journals. I frequently serve as a PC member or reviewer of various prestigious conferences such as ICSE (2020, 2022, 2023), ASE (2019, 2020), ISSTA (2019, 2020), and journals (various years) such as TSE, TOSEM, TNSM.

Chairs. Apart from serving as PC member/reviewer, I have also been invited to serve as chair for specialized tracks at various conferences. Particularly, I served as chair for Student Research Competition Track at ASE'22, Demonstration Track at IBC'22, and Software Engineering Track at SoICT'23.

Expert panel. I was invited to serve on a panel of world-leading researchers on Automated Bug Fixing to discuss future directions for the research area (organized with ICSE 2023 in Melbourne).

Overall, the opportunities to serve as chairs or expert panel, aside from PC member/reviewer, are great pathways for my research career to aim for bigger goals in the future and to be recognized as leading researcher in the field of research that I am pursuing.